DCTC-Net: Dual-Branch Cross-Fusion Transformer–CNN Architecture for Medical Image Segmentation

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***Abstract*—Hybrid architectures that combine convolutional neural networks (CNNs) with Transformers have emerged as a promising approach for medical image segmentation. However, existing networks based on this hybrid architecture often encounter two challenges. First, while the CNN branch effectively captures local image features through convolution operations, vanilla convolution lacks the ability to achieve adaptive feature extraction. Second, although the Transformer branch can model global image information, conventional self-attention (SA) primarily focuses on spatial relationships, neglecting channel and cross-dimensional attention, leading to suboptimal segmentation results, particularly for medical images with complex backgrounds. To address these limitations, we propose a dual-branch cross-fusion Transformer–CNN architecture for medical image segmentation (DCTC-Net). Our network provides two key advantages. First, a dynamic deformable convolution (DDConv) is integrated into the CNN branch to overcome the limitations of adaptive feature extraction with fixed-size convolution kernels and also eliminate the issue of shared convolution kernel parameters across different inputs, significantly enhancing the feature expression capabilities of the CNN branch. Second, a (shifted)-window adaptive complementary attention module ((S)W-ACAM) and compact convolutional projection are incorporated into the Transformer branch, enabling the network to comprehensively learn cross-dimensional long-range dependencies in medical images. Experimental results demonstrate that the proposed DCTC-Net achieves superior medical image segmentation performance compared to state-of-the-art (SOTA) methods, including CNN and Transformer networks. In addition, our DCTC-Net requires fewer parameters and lower computational costs and does not rely on pretraining.**

***Index Terms*—Deep learning, dynamic deformable convolution (DDConv), medical image segmentation, (shifted)-window adaptive complementary attention.**

I. Introduction

MEDICAL image segmentation involves partitioning a medical image into distinct regions with unique characteristics. Accurate medical image segmentation can not only facilitate abnormal detection in human body regions but also guide clinicians. Consequently, precise medical image segmentation has become a key component of computer-aided diagnosis, patient condition analysis, image-guided surgery, tissue and organ reconstruction, and treatment planning. Compared to natural RGB images, medical images often exhibit high-density noise, low contrast, and blurred edges. Thus, the challenge of rapidly and accurately segmenting specific human organs and lesions from medical images remains a major hurdle in the field of smart medicine.

Traditional methods for medical image segmentation initially relied on handcrafted features developed by medical experts based on their domain expertise [3]. These methods, while possessing a strong mathematical basis and theoretical support, often suffer from poor generalization abilities when applied to different organs or lesions in the human body. Later, inspired by fully convolutional networks (FCNs) [4] and the encoder–decoder paradigm, Ronneberger et al. [1] designed the U-Net, which was first applied to medical image segmentation. Since then, the U-shaped encoder–decoder structure has garnered widespread attention. The small parameter count and effective segmentation performance of the U-Net have enabled deep learning to achieve breakthroughs in the field of medical image segmentation. Subsequently, a variety of improved medical image segmentation networks have been developed, such as 2-D U-Net++ [5], ResDO-UNet [6], SGU-Net [7], 2.5-D RIU-Net [8], 3-D Unet [9], and V-Net [10]. The rapid advancement of convolutional neural networks (CNNs) in medical image segmentation is mainly attributable to the scale invariance and inductive bias inherent in the convolution operation. However, the fixed receptive field limits CNNs' ability to effectively capture long-range spatial dependencies among pixels in medical images.

To address the limitations of CNNs in capturing global dependencies, Vaswani et al. [11] proposed the Transformer architecture originally for natural language processing (NLP), which was later adapted for image analysis. By leveraging self-attention (SA) mechanisms and long-range dependence modeling, Transformers effectively overcome the problem of CNNs being limited to local feature extraction. Numerous Transformer-based methods have been applied to medical image segmentation, including Swin-Unet [12], BAT [13], Swin UNETR [14], and UCTransNet [15]. These methods can be broadly categorized into pure Transformer architectures and hybrid CNN-Transformer architectures. Pure Transformer architectures rely on SA to model long-range dependencies. However, due to the lack of inductive bias in the Transformer

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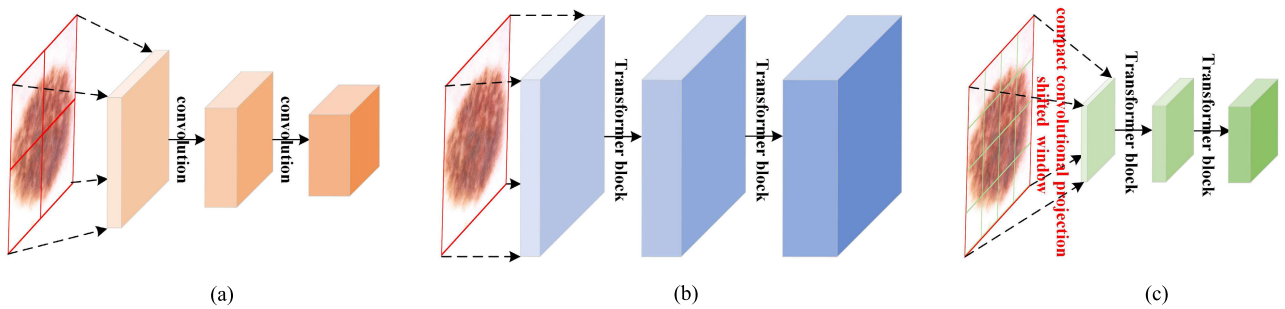


Fig. 1. Comparisons of different architectures. (a) CNN-based medical image segmentation network [1] usually uses the pyramid structure of convolution operations to intensively predict medical image features, and the limited receptive field becomes larger with the deepening of network layers. (b) Transformer structure [2] uses the global receptive field to intensively predict medical image features. Although it can effectively capture the global features of images, it requires a large number of calculations and parameters. (c) Transformer branch of DCTC-Net uses the shifted window and compact convolutional projection to effectively reduce the number of parameters and computational costs while realizing the capture of global features of medical images.

architecture itself, Transformers perform poorly on small-scale datasets [16]. Moreover, Transformers tend to neglect local detailed features, which reduces the separability between the background and the foreground, particularly for small lesions or objects with significant scale variations. Hybrid CNN-Transformer architectures leverage the complementary strengths of both CNNs and Transformers to model both local and global information in medical images, achieving promising results. However, these hybrid architectures still suffer from two main limitations. First, these networks often neglect organ deformation and lesion irregularities when modeling local features, resulting in limited representation of local details. Second, these networks often ignore the correlation between spatial and channel dimensions when modeling global features, resulting in inadequate SA expression. To address the aforementioned problems, our main contributions are as follows.

1) We propose a novel dynamic deformable convolution (DDConv) module for medical image feature extraction. Leveraging task-adaptive learning, DDConv flexibly adjusts both the weight coefficients and deformation offsets of the convolution operation adaptively. DDConv overcomes the limitations of fixed, narrow receptive fields and nonadaptive convolution kernel parameters that are prevalent in vanilla convolution and its variants, such as dilated convolution and involution. Consequently, DDConv enhances the network's ability to detect subtle lesions and targets exhibiting significant scale variations in medical images.
2) We introduce a novel (shifted)-window adaptive complementary attention module ((S)W-ACAM). The module enables cross-dimensional global modeling of medical images through four parallel branches using adaptive weight coefficient learning. In contrast to existing attention mechanisms, such as CBAM and Nonlocal, (S)W-ACAM effectively addresses the limitations of conventional attention by explicitly modeling the cross-dimensional relationships between spatial features and channels. This allows (S)W-ACAM to capture cross-dimensional, long-range correlations within medical images and enhance the separability between segmented objects and the background. As illustrated in Fig. 1, by integrating shifted windows and compact convolutional projection, our (S)W-ACAM significantly reduces the network's parameter count and computational complexity compared to conventional CNNs and pure Transformer architectures while maintaining the capability to capture comprehensive global features.
3) We present a novel parallel network structure for medical image segmentation, termed dual-branch cross-fusion Transformer-CNN architecture for medical image segmentation (DCTC-Net), based on adaptive CNNs and cross-dimensional Transformer architectures. In contrast to existing hybrid architectures that combine CNNs with Transformers, such as Swin-Unet [12] and Swin UNETR [14], DCTC-Net enhances representation learning by effectively integrating local and global features at various resolutions through cross-fusion between CNNs and Transformers. Moreover, DCTC-Net eliminates the need for pretraining, requires fewer parameters, and incurs lower computational costs, with only 11.58 M parameters and 4.53 GFLOPs.

II. Related Work

Medical image segmentation plays a crucial role in the field of medical image processing and is also one of the key technologies of computer-aided diagnosis and treatment systems. In recent years, with the rapid development of deep learning techniques, researchers have continuously developed numerous deep network models for medical image segmentation. Generally, medical image segmentation methods can be coarsely classified into two categories: CNN-based methods and Transformer-based methods.

A. CNNs-Based Methods

CNNs have demonstrated remarkable success in medical image segmentation by learning hierarchical feature representations through multilayer architectures. Unlike traditional handcrafted methods, CNN-based approaches can automatically extract discriminative features from large-scale medical datasets, making them indispensable components of modern intelligent medical systems.

Ronneberger et al. [1] drew inspiration from the FCN [4] and designed the first end-to-end network, U-Net, for medical image segmentation in the ISBI cell tracking challenge. The

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U-Net architecture employs a symmetrical encoder–decoder structure, enabling the full utilization of local contextual information within medical images while reducing the reliance on extensive training datasets. Consequently, U-Net can achieve effective medical image segmentation even with limited data. Building upon U-Net, Alom et al. [17] developed R2U-Net by integrating U-Net with ResNet [18] and recurrent neural network (RCNN) components [19], achieving strong performance across multiple medical image segmentation datasets, including blood vessels and retinas. Subsequently, Gu et al. [21] introduced dynamic convolution [20] into U-Net, proposing CA-Net. Experiments using medical datasets have shown that CA-Net not only improves the segmentation accuracy of medical images but also reduces the network training time. Building on U-Net, Yang et al. [23] incorporated the idea of residual connections and deformable convolution [22], integrating residual deformable convolution into U-Net and proposing DCU-Net. DCU-Net achieves superior segmentation performance compared to U-Net on the DRIVE medical dataset. Lei et al. [7] designed SGU-Net based on U-Net, proposing an ultralight convolution module and an additional adversarial shape-constraint that significantly improve the segmentation accuracy of abdominal medical images through self-supervised training. Although CNNs have made significant progress in architecture design, their success is primarily due to scale invariance and inductive bias in local modeling. However, fixed receptive fields limit their ability to capture relationships between distant pixels in medical images and constrain the capacity to model long-range dependencies.

Compared to existing approaches, such as vanilla convolution [1], dynamic convolution [20], and deformable convolution [22], our DDConv not only adaptively adjusts the weight coefficient and deformation offsets of the convolution kernels according to the medical image characteristics, but also better adapts to the shape of organs and small lesions with significant scale variations. Consequently, it enhances the local feature expression ability of the segmentation network.

B. Transformer-Based Methods

Vaswani et al. [11] introduced the Transformer. Due to its unique architecture, Transformers gained the ability to process inputs of variable length, establish long-range dependence modeling, and capture global information. Transformer's success is primarily attributed to the SA mechanism, enabling it to capture long-range dependencies.

Building on the Transformer's excellent performance in NLP, ViT [2] represents the first successful application of Transformers in image processing, capturing the global context information of input images through multiple cascaded Transformer layers. This contributes to the successful adoption of Transformers in image classification. Subsequently, Chen et al. [24] introduced TransUNet, marking a new era for Transformers in medical image segmentation. Since TransUNet directly adopts the Transformer network from NLP for image segmentation, the input image patch size is fixed, resulting in high computational overhead. To address this limitation, Valanarasu et al. [25] proposed MedT, which incorporates a gating mechanism that can be automatically adjusted to obtain position embedding weights suitable for datasets of varying sizes. Recognizing the greater complexity and higher resolution of images compared to text, Cao et al. [12] proposed Swin-Unet, a pure Transformer network for medical image segmentation that incorporates the shifted window multihead SA (SW-MSA) mechanism from Swin Transformer [26]. Swin-Unet achieved state-of-the-art (SOTA) segmentation performance on the Synapse and ACDC multiorgan segmentation datasets upon its release. To better leverage Transformers for processing dermoscopic image data, Wang et al. [13] designed BAT, a network based on boundary detection principles. The proposed boundary-wise attention gate (BAG) fully utilizes prior knowledge of image boundaries to capture finer details in medical images. BAT achieved impressive Dice scores on skin lesion datasets, surpassing many existing medical image segmentation networks.

In contrast to the SA mechanisms found in existing Transformer architectures [12], [13], our (S)W-ACAM requires fewer parameters and lower computational cost while effectively capturing global cross-dimensional long-range dependencies in medical images and enhancing the global feature expression capability of the segmentation network.

Our DCTC-Net, while eliminating the need for pretraining on large labeled datasets, effectively preserves local details and global semantic information in medical images. Specifically, DCTC-Net leverages DDConv to capture complex local features often obscured by noise or variability, while (S)W-ACAM leverages global context to improve segmentation accuracy and robustness across different anatomical structures. These benefits are demonstrated in the ablation study and through competitive quantitative results on the ISIC2018, LiTS-Liver, and ACDC datasets.

III. Method

A. Overall Architecture

The fusion of local and global features significantly enhances performance in medical image segmentation. CNNs capture local features within medical images through convolution operations and hierarchical feature representation. Conversely, Transformers achieve the extraction of global features in medical images through cascaded SA mechanisms and contextual matrix operations. To fully leverage both local details and global semantic features present in medical images, we propose a parallel interactive network architecture: DCTC-Net. The overall architecture of the network is illustrated in Fig. 2(a). DCTC-Net capitalizes on the complementary properties of CNNs and Transformers. During forward propagation, DCTC-Net continuously feeds local details extracted by the CNN branch to the decoder of the Transformer branch. Similarly, DCTC-Net also feeds the global long-range relationships captured by the Transformer branch to the decoder of the CNN branch. Consequently, the proposed DCTC-Net provides superior local and global feature representation compared to pure CNN or Transformer networks, showcasing significant potential in the field of medical image segmentation.

Specifically, DCTC-Net comprises four key components: a patch embedding module, a dynamically adaptive CNN

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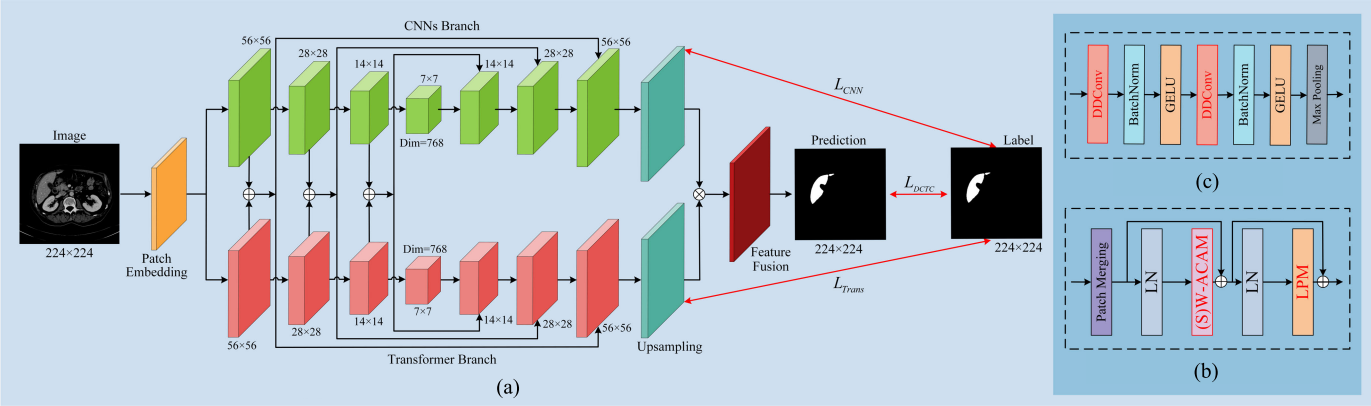


Fig. 2. (a) Architecture of DCTC-Net. DCTC-Net consists of a dual-branch interaction between dynamically adaptive CNNs and cross-dimensional feature fusion Transformer. The DDConv in the CNN branch can adaptively adjust the weight coefficient and deformation offset of the convolution operation, which improves the segmentation accuracy of irregular objects in medical images. The (S)W-ACAM in the Transformer branch can capture the cross-dimensional long-range dependencies in medical images, improving the separability of segmented objects and backgrounds in medical images. The LPM significantly reduces the parameters and computational cost of the original Transformer network by using the Ghost strategy. (b) Two successive Transformer blocks. W-ACAM and SW-ACAM are cross-dimensional SA modules with shifted windows and compact convolutional projection configurations.

branch, a cross-dimensional fusion Transformer branch, and a feature fusion module. The design of the dynamically adaptive CNN branch and the cross-dimensional fusion Transformer branch draws inspiration from the U-Net and Swin-Unet architectures, respectively. The dynamically adaptive CNN branch consists of seven primary stages. A central challenge in medical image segmentation is effectively handling multiscale change targets, which refer to the significant variations in size, shape, and morphology exhibited by segmentation targets (such as organs or lesions) across different images or even within the same image. Traditional convolution operations, employing fixed-size kernels and static receptive fields, struggle to adapt to this inherent variability. To overcome this limitation, our CNN branch utilizes the dynamically adaptive DDConv in each stage. Unlike traditional fixed kernels, DDConv dynamically learns both deformation offsets and weight coefficients based on the input features. The learned offsets allow the receptive field's shape and sampling locations to adapt to the specific geometry of organs or lesions, while the learned weight coefficients adjust the feature emphasis dynamically. This dual adaptivity makes DDConv particularly adept at capturing features across diverse scales and complex shapes, enabling the network to better understand local semantic features, perceive subtle changes, and accurately extract these multiscale change targets. Similarly, the cross-dimensional fusion Transformer branch also consists of seven main stages. By employing (S)W-ACAM attention in each stage, as illustrated in Fig. 2(b), the segmentation network gains enhanced understanding of the global dependencies within medical images. This allows it to effectively capture positional information between different organs and improve the separability between foreground and background.

Although our DCTC-Net can effectively improve the feature representation of medical images, the dual-branch structure requires a substantial amount of training data and network parameters. The conventional Transformer network contains numerous multilayer perceptron (MLP) layers, which not only increase the training burden of the network but also cause the number of model parameters to rise sharply, resulting in slower model training. Inspired by the Ghost network [27], we redesigned the MLP layer in the original Transformer and proposed a lightweight perceptron module (LPM). The LPM enables our DCTC-Net to not only achieve better medical image segmentation results than using MLP layers but also

TABLE I

VARIANTS OF DCTC-NET FOR MEDICAL IMAGE SEGMENTATION. THE EXAMPLE INPUT SIZE IS 224 × 224. *S* REPRESENTS THE FEATURE MAPS' HEIGHT (AND WIDTH) OF EACH STAGE. *D* AND *H* MEAN EMBEDDING DIMENSIONS AND THE NUMBER OF HEADS IN THE (S)W-ACAM MODULE, RESPECTIVELY. *d* REPRESENTS THE DIMENSION OF *Q*, *K* AND *V* AFTER COMPACT CONSTRUCTIVE PROJECTION

		Output Size	DCTC-Net-T	DCTC-Net-B
Encoder	Stage1	$\begin{bmatrix} 56 \times 56 \\ (S_1 = 56) \end{bmatrix}$	$\begin{bmatrix} D_1 = 96 \\ H_1 = 3 \\ d_1 = 12 \end{bmatrix} \times 2$	$\begin{bmatrix} D_1 = 96 \\ H_1 = 4 \\ d_1 = 12 \end{bmatrix} \times 2$
	Stage2	$\begin{bmatrix} 28 \times 28 \\ (S_2 = 28) \end{bmatrix}$	$\begin{bmatrix} D_2 = 192 \\ H_2 = 6 \\ d_2 = 24 \end{bmatrix} \times 2$	$\begin{bmatrix} D_2 = 192 \\ H_2 = 8 \\ d_2 = 24 \end{bmatrix} \times 2$
	Stage3	$\begin{bmatrix} 14 \times 14 \\ (S_3 = 14) \end{bmatrix}$	$\begin{bmatrix} D_3 = 384 \\ H_3 = 12 \\ d_3 = 48 \end{bmatrix} \times 6$	$\begin{bmatrix} D_3 = 384 \\ H_3 = 16 \\ d_3 = 48 \end{bmatrix} \times 18$
	Stage4	$\begin{bmatrix} 7 \times 7 \\ (S_4 = 7) \end{bmatrix}$	$\begin{bmatrix} D_4 = 768 \\ H_4 = 24 \\ d_4 = 96 \end{bmatrix} \times 2$	$\begin{bmatrix} D_4 = 768 \\ H_4 = 32 \\ d_4 = 96 \end{bmatrix} \times 2$
Decoder	Stage5	$\begin{bmatrix} 14 \times 14 \\ (S_5 = 14) \end{bmatrix}$	$\begin{bmatrix} D_5 = 384 \\ H_5 = 12 \\ d_5 = 48 \end{bmatrix} \times 6$	$\begin{bmatrix} D_5 = 384 \\ H_5 = 16 \\ d_5 = 48 \end{bmatrix} \times 18$
	Stage6	$\begin{bmatrix} 28 \times 28 \\ (S_6 = 28) \end{bmatrix}$	$\begin{bmatrix} D_6 = 192 \\ H_6 = 6 \\ d_6 = 24 \end{bmatrix} \times 2$	$\begin{bmatrix} D_6 = 192 \\ H_6 = 8 \\ d_6 = 24 \end{bmatrix} \times 2$
	Stage7	$\begin{bmatrix} 56 \times 56 \\ (S_7 = 56) \end{bmatrix}$	$\begin{bmatrix} D_7 = 96 \\ H_7 = 3 \\ d_7 = 12 \end{bmatrix} \times 2$	$\begin{bmatrix} D_7 = 96 \\ H_7 = 4 \\ d_7 = 12 \end{bmatrix} \times 2$
	Para. (M)		11.58	21.24
	GFLOPs		4.53	13.29

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TABLE II

PERFORMANCE COMPARISON OF THE PROPOSED METHOD AGAINST THE SOTA APPROACHES ON THE ISIC2018 BENCHMARKS. RED INDICATES THE BEST RESULT, AND BLUE DISPLAYS THE SECOND BEST

Method		DI↑	JA↑	SE↑	AC↑	SP↑	Para. (M) ↓	GFLOPs
CNNs	U-Net [1]	86.54	79.31	88.56	93.16	96.44	34.52	65.39
	R2UNet [17]	87.92	80.28	90.92	93.38	96.33	39.09	152.82
	Attention Unet [31]	87.16	79.55	88.52	93.17	95.62	34.88	66.57
	CENet [32]	87.61	81.18	90.71	94.03	96.35	29.02	11.79
	CPFNet † [33]	90.18	82.92	91.66	94.68	96.63	30.65	9.15
	SSNet [34]	90.69	83.95	**92.71**	94.91	96.78	47.51	197.26
Transformer	Swin-Unet † [12]	89.26	80.47	90.36	94.45	96.51	41.40	11.63
	TransUNet † [35]	89.39	82.10	91.43	93.67	96.54	105.30	15.21
	BAT † [13]	90.21	83.49	91.59	94.85	96.57	45.56	13.38
	CvT † [36]	88.23	80.21	87.60	93.68	96.28	21.51	20.53
	PVT [37]	87.31	79.99	87.74	93.10	96.21	28.86	14.92
	CrossForm [38]	87.44	80.06	88.25	93.39	96.40	38.66	13.57
	BRAU-Net++ [39]	90.08	83.87	91.06	95.18	96.27	50.76	22.39
	HResFormer [40]	90.37	84.15	91.33	94.97	96.51	117.45	131.72
	MCAFT-Net [41]	90.56	84.35	91.57	94.53	96.44	35.43	15.48
	CSWin-UNet [42]	90.61	**84.62**	91.83	95.16	**97.37**	23.57	**4.72**
Mamba	U-Mamba [43]	87.54	78.78	90.15	94.26	96.07	101.27	453.89
	Mamba-UNet [44]	83.44	77.86	89.74	93.74	96.02	28.00	7.81
	VM-Unet [45]	89.77	81.42	91.18	94.95	96.16	44.27	7.56
	UD-Mamba [46]	89.16	81.96	89.57	94.63	96.24	35.18	19.74
	DMA-Net [47]	89.53	81.79	90.62	94.71	96.26	26.47	15.76
	TMU † [48]	89.18	81.21	91.87	94.58	96.48	27.46	21.87
DCTC-Net-T (our)		**90.72**	84.59	92.54	**95.21**	96.83	**11.58**	**4.53**
DCTC-Net-B (our)		**91.23**	**84.76**	**92.68**	**95.56**	**98.21**	**21.24**	13.29

† indicates the model is initialized with pre-trained weights on the ImageNet21K. "Para." refers to the number of parameters. "GFLOPs" is calculated with the input resolution of 224×224. Since the dermoscopic images are 2D medical images, the comparison methods are all 2D networks.

TABLE III

PERFORMANCE COMPARISON OF THE PROPOSED METHOD AGAINST THE SOTA APPROACHES ON THE LITS-LIVER BENCHMARKS. RED INDICATES THE BEST RESULT, AND BLUE DISPLAYS THE SECOND BEST

Method		DI ↑	VOE ↓	RVD ↓	ASD ↓	RMSD ↓	Para. (M) ↓	GFLOPs
CNNs	U-Net [1]	93.99±1.23	11.13±2.47	3.22±0.20	5.79±0.53	123.57±6.28	34.52	65.39
	R2UNet [17]	94.01±1.18	11.12±2.37	2.36±0.15	5.23±0.45	120.36±5.03	39.09	152.82
	Attention Unet [31]	94.08±1.21	10.95±2.36	3.02±0.18	4.95±0.48	118.67±5.31	34.88	66.57
	CENet [32]	94.04±1.15	11.03±2.31	6.19±0.16	4.11±0.51	115.40±5.82	29.02	11.79
	3D Unet [9]	94.10±1.06	11.13±2.23	1.42±0.13	2.61±0.45	36.43±5.38	40.32	66.45
	V-Net [10]	94.25±1.03	10.65±2.17	1.92±0.11	2.48±0.38	38.28±5.05	65.17	55.35
	SSNet [34]	96.37±1.16	9.58±2.13	**1.41±0.11**	2.29±0.36	36.35±4.98	47.51	197.26
Transformer	Swin-Unet † [12]	95.62±1.32	9.73±2.16	2.78±0.21	2.35±0.35	38.85±5.42	41.40	11.63
	TransUNet † [35]	95.79±1.09	9.82±2.10	1.98±0.15	2.33±0.41	37.22±5.23	105.30	15.21
	CvT † [36]	95.81±1.25	9.66±2.31	1.77±0.16	2.34±0.29	36.71±5.09	21.51	20.53
	PVT [37]	94.56±1.15	9.75±2.19	1.69±0.12	2.42±0.34	37.35±5.16	28.86	14.92
	CrossForm [38]	94.63±1.24	9.72±2.24	1.65±0.15	2.39±0.31	37.21±5.32	38.66	13.57
	BRAU-Net++ [39]	95.87±1.35	9.63±2.15	1.52±0.18	2.34±0.36	36.67±5.59	50.76	22.39
	HResFormer [40]	96.33±1.23	9.55±2.13	1.50±0.13	**2.19±0.42**	36.47±5.16	117.45	131.72
	MCAFT-Net [41]	96.41±1.08	9.63±2.24	1.47±0.11	2.30±0.37	36.29±5.04	35.43	15.48
	CSWin-UNet [42]	96.45±1.16	**9.51±2.16**	1.43±0.16	2.29±0.35	**36.12±4.99**	23.57	**4.72**
Mamba	U-Mamba [43]	94.96±1.22	9.70±2.11	1.57±0.23	2.37±0.31	37.08±6.67	101.27	453.89
	Mamba-UNet [44]	95.34±1.29	9.67±2.24	1.56±0.17	2.36±0.29	36.86±5.97	28.00	7.81
	VM-Unet [45]	95.93±1.17	9.59±2.21	1.53±0.14	2.28±0.34	36.51±4.99	44.27	7.56
	UD-Mamba [46]	95.68±1.08	9.63±2.14	1.55±0.19	2.36±0.32	36.62±5.24	35.18	19.74
	DMA-Net [47]	96.01±1.27	9.61±2.25	1.52±0.16	2.32±0.34	36.48±5.16	26.47	15.76
	TMU † [48]	95.98±1.16	9.60±2.16	1.51±0.13	2.31±0.38	36.52±4.96	27.46	21.87
DCTC-Net-T (our)		**96.48±1.05**	9.53±2.11	1.45±0.12	2.29±0.33	36.21±4.97	**11.58**	**4.53**
DCTC-Net-B (our)		**96.82±1.22**	**9.46±2.33**	**1.38±0.13**	**2.21±0.35**	**36.08±4.88**	**21.24**	13.29

† indicates the model initialized is with pre-trained weights on ImageNet21K. "Para." refers to the number of parameters. "GFLOPs" is calculated with the input resolution of 224×224. Compared with the comparison experiment oin the ISIC2018 dataset, 3D Unet and V-Net are introduced into the comparison experiment on the LiTS-Liver dataset.

significantly reduce the number of parameters and computational costs; the Transformer can even achieve good results without a large amount of labeled data for training. Notably, the dual-branch structure involves symmetric encoders and

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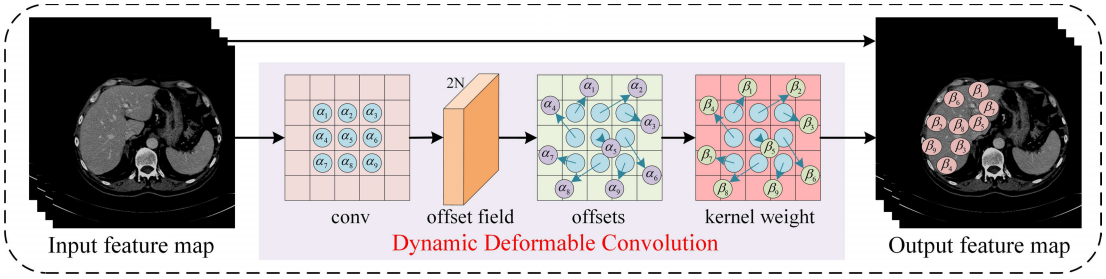


Fig. 3. Module of the proposed DDConv. Compared with existing popular convolution strategies, DDConv can dynamically adjust the weight coefficient and deformation offset of the convolution operation during the training process, which facilitates feature capture and extraction of irregular targets in medical images. α and β represent the different weight values of DDConv in different states.

decoders, allowing the parallel interaction network structure to maximize the preservation of both local features and global features in medical images.

While the DCTC-Net architecture's dual-branch design could potentially increase computational complexity, strategic optimizations—including the efficient DDConv in the CNN branch, as well as (S)W-ACAM and LPM within the Transformer branch—effectively mitigate this concern. As shown in the Experiment and Results section, DCTC-Net achieves highly competitive GFLOPs compared with SOTA methods. We have leveraged the core framework of our previous work [28], and further enhanced the generalization and interpretability of the method.

B. Dynamic Deformable Convolution

Vanilla convolution exhibits spatial invariance and channel specificity, which limits its capacity to adapt to varying visual patterns across different spatial locations. Moreover, the fixed receptive field of vanilla convolution makes it challenging to effectively extract features from small targets or those with blurred edges. Consequently, vanilla convolution inherently possesses limited adaptability and weak generalization ability for complex feature representation in medical images. While existing deformable convolution [22] and dynamic convolution approaches [20] outperform vanilla convolution to some extent, they still face the problem of balancing network performance and model size when applied to medical image segmentation.

To overcome the aforementioned shortcomings of current convolution operations, this article introduces a novel convolution strategy termed DDConv, illustrated in Fig. 3. As shown, DDConv adaptively learns kernel deformation offsets and weight coefficients based on the specific task and data distribution, thereby changing both the shapes and values of convolution kernels. It can effectively address the problems of large data distribution differences and significant target deformation encountered in medical image segmentation. Furthermore, DDConv is designed as a plug-and-play module, readily embeddable within any network structure.

The shape transformation of the convolutional kernel in DDConv is based on the network's learning of deformation offsets. The segmentation network initially samples the input feature map X using a square convolutional kernel S, and then performs a weighted summation with a weight matrix M. The square convolution kernel S defines the extent of the receptive field; for example, a 3×3 convolution kernel can be expressed as

$$S = \{(0,0),(0,1),(0,2),\ldots,(2,1),(2,2)\}. \tag{1}$$

Then, the output feature map Y at the coordinate φ_n can be expressed as

$$Y(\varphi_n) = \sum_{\varphi_m \in S} W \cdot X(\varphi_n + \varphi_m) \tag{2}$$

where W is the convolutional kernel weight matrix. When the deformation offset $\triangle\varphi_m = \{m = 1,2,3,\ldots,N\}$ is introduced in the weight matrix M, N is the total length of S. Thus (2) can be expressed as

$$Y(\varphi_n) = \sum_{\varphi_m \in S} W \cdot X(\varphi_n + \varphi_m + \triangle\varphi_m). \tag{3}$$

Through network learning, an offset matrix with the same size as the input feature map can be obtained, and the matrix dimension is twice that of the input feature map.

To demonstrate that the convolution kernel of DDConv is dynamic, we first present the output feature map of vanilla convolution

$$y = \sigma(W \cdot X) \tag{4}$$

where σ is the activation function, and y is the output feature map. In contrast, the output feature map of DDConv is

$$\hat{y} = \sigma((\alpha_1 \cdot W_1 + \cdots + \alpha_n \cdot W_n) \cdot X) \tag{5}$$

where n is the number of weight coefficients, α_n is the weight coefficient with learnable parameters, and $\hat{y}$ is the output feature map generated by DDConv. DDConv dynamically adjusts the convolution kernel weights by linearly combining different weight matrices according to the corresponding weight coefficients before performing the convolution operation.

Based on the preceding analysis, it is evident that DDConv dynamically adjusts both the shape and weights of the convolution kernel. In contrast to directly increasing the number and size of convolution kernels, DDConv offers a simpler and more efficient alternative. The proposed DDConv not only addresses the limited adaptive feature extraction capability of fixed-size convolution kernels but also mitigates the inherent drawback of shared convolution kernel parameters across different inputs. As a result, our DDConv can effectively enhance the segmentation accuracy of both small targets and large targets with blurred boundaries in medical images.

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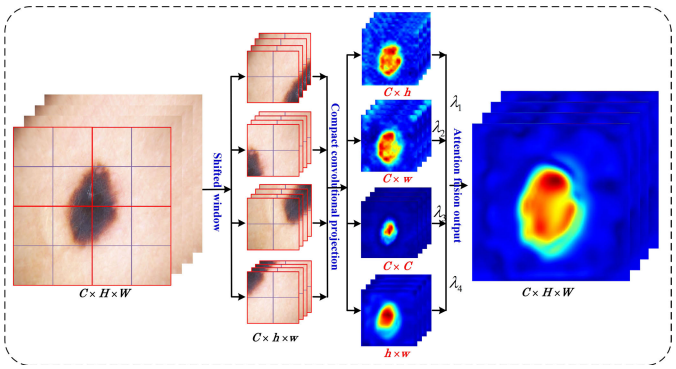


Fig. 4. Module of the proposed (S)W-ACAM. Unlike conventional SA, (S)W-ACAM has the advantages of spatial and channel attention, and can also capture long-distance correlation features between spatial and channel dimensions. Through the shifted window operation, the spatial resolution of images is significantly reduced, and through the compact convolutional projection operation, the channel dimension of images is also significantly reduced. Thus, the overall computational costs and complexity of our proposed network are reduced. λ_1–λ_4 are learnable weight parameters.

C. Shifted Window Adaptive Complementary Attention Module

The SA mechanism, a core component of Transformers, captures long-range dependencies within feature maps through matrix operations. However, SA mechanisms primarily focus on dependencies in the spatial dimension, neglecting cross-dimensional dependencies between spatial and channel features [29]. As a result, when applied to medical image segmentation characterized by low-contrast and high-density noise, SA mechanisms may struggle to distinguish segmentation targets from the background, leading to suboptimal performance.

To address the aforementioned limitations, we introduce a novel cross-dimensional SA module termed (S)W-ACAM. As illustrated in Fig. 4, (S)W-ACAM consists of four parallel branches: the top two branches implement a conventional dual attention module, while the bottom two branches are cross-dimensional attention modules. Compared to existing SA modules, such as spatial SA, channel SA, and dual SA, our proposed (S)W-ACAM can not only effectively extract long-range dependencies in both spatial and channel dimensions, but also capture the cross-dimensional long-range dependencies between spatial and channel features. The four branches are designed to be complementary and enhance the separability between foreground and background. Therefore, the four branches together provide richer long-range dependence relationships, ultimately improving segmentation results for medical images.

The standard Transformer architecture [2] employs global SA, calculating the relationship between one token and all other tokens, resulting in high computational cost. This complexity increases exponentially with image size. To improve computational efficiency, we adopt the shifted window calculation method, inspired by Swin Transformer [26], which restricts SA calculations to local windows. However, directly applying the shifted window method to our four-branch (S)W-ACAM module does not sufficiently reduce the overall computational complexity. Therefore, we also introduce compact convolutional projection. First, we reduce the spatial size of the medical image through the shifted window operation. Second, we compress the channel dimension of feature maps using compact convolutional projection. Finally, we perform the SA calculation. Notably, this method not only better captures global high-dimensional information in medical images but also significantly reduces the computational costs. With an image containing $h \times w$ patches, each within windows of size $M \times M$, the complexity of the (S)W-ACAM, global MSA in the original Transformer, and the (S)W-MSA in Swin Transformer are compared as follows:

$$\Omega\left(\text{MSA}\right) = 4hwC^2 + 2\left(hw\right)^2 C \tag{6}$$

$$\Omega\left(\left(\text{S}\right)\text{W-MSA}\right) = 4hwC^2 + 2M^2hwC \tag{7}$$

$$\Omega\left(\left(\text{S}\right)\text{W-ACAM}\right) = \frac{hwC^2}{4} + M^2hwC. \tag{8}$$

The former term of each formula is a quadratic function of the number of patches hw, while the latter term is linear when M is fixed (the default is 7). Then, the computational cost of (S)W-ACAM is smaller than MSA and (S)W-MSA.

Among the four parallel branches of (S)W-ACAM, two branches are used to capture channel correlation and spatial correlation, and the remaining two branches are used to capture the correlation between channel dimension C and spatial dimension H, and between channel dimension C and spatial dimension W. After adopting the shifted window partitioning method, as shown in Fig. 2(b), the calculation process of successive Transformer blocks is as follows:

$$\hat{T}^l = \text{W-ACAM}\left(\text{LN}\left(T^{l-1}\right)\right) + T^{l-1} \tag{9}$$

$$T^l = \text{LPM}\left(\text{LN}\left(\hat{T}^l\right)\right) + \hat{T}^l \tag{10}$$

$$\hat{T}^{l+1} = \text{SW-ACAM}\left(\text{LN}\left(T^l\right)\right) + T^l \tag{11}$$

$$T^{l+1} = \text{LPM}\left(\text{LN}\left(\hat{T}^{l+1}\right)\right) + \hat{T}^{l+1} \tag{12}$$

where $\hat{T}^l$ and T^l represent the output features of (S)W-ACAM and LPM, respectively. W-ACAM represents window adaptive complementary attention, SW-ACAM represents shifted window adaptive complementary attention, and LPM represents the lightweight perceptron module. For the specific attention calculation process of each branch, we follow the same principle as Swin Transformer, as follows:

$$\text{Attention}\left(Q, K, V\right) = \text{SoftMax}\left(\frac{QK^{\text{T}}}{\sqrt{C/8}} + B\right)V \tag{13}$$

where relative position bias $B \in \mathbb{R}^{N_{\text{patch}} \times N_{\text{patch}}}$, and $Q, K, V \in \mathbb{R}^{N_{\text{patch}} \times C/8}$ are query, key, and value matrices, respectively. $C/8$ represents the dimension of query/key, and N_{patch} represents the number of patches.

After four parallel attention branches Out_1–Out_4 are calculated, the final feature fusion output is

$$\text{Out} = \lambda_1 \cdot \text{Out}_1 + \lambda_2 \cdot \text{Out}_2 + \lambda_3 \cdot \text{Out}_3 + \lambda_4 \cdot \text{Out}_4 \tag{14}$$

where λ_1–λ_4 are learnable parameters that enable adaptive control of the importance of each attention branch.

Unlike other SA mechanisms, the (S)W-ACAM presented in this article comprehensively captures the correlation between

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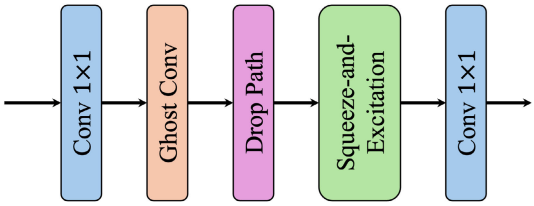


Fig. 5. Module of the proposed LPM. The LPM comprises a linear dimensionality reduction layer (Conv 1 × 1), a Ghost Conv module for efficient feature expansion, a drop path layer for regularization, a squeeze-and-excitation (SE) block for channel attention, and a linear dimensionality expansion layer.

spatial and channel dimensions and effectively leverages the contextual information present in medical images to enable long-range dependence modeling. By effectively addressing the limitation of conventional SA—namely, its primary focus on spatial SA while neglecting channel and cross-dimensional SA—our (S)W-ACAM achieves superior feature representation for medical images characterized by high noise levels, low contrast, and complex backgrounds.

D. Lightweight Perceptron Module

To reduce the number of parameters and computational complexity of the Transformer branch while preserving its ability to capture long-range dependencies, we adopted principles from GhostNet [27] to redesign the MLP layer in the Transformer, proposing an LPM, whose structure is shown in Fig. 5. The LPM primarily consists of five parts: a linear dimensionality reduction layer, a Ghost module, a drop path, a channel attention, and a linear dimensionality expansion layer.

To reduce computational costs, the LPM first performs a linear transformation on the input data to reduce the feature dimensionality. The linear dimensionality reduction is implemented using a 1×1 convolution, reducing the number of input channels from C to C/r, where r is the reduction ratio used to control the extent of reduction. This operation can be expressed as

$$X' = \mathrm{ReLU}\left(\mathrm{Conv}_{1\times1}(X)\right) \tag{15}$$

where $X \in \mathbb{R}^{H\times W\times C}$ is the input feature tensor, $X' \in \mathbb{R}^{H\times W\times C/r}$ is the dimensionality-reduced feature tensor, and $\mathrm{Conv}_{1\times1}(\cdot)$ represents the 1×1 convolution operation.

After reducing the number of channels, we utilize the Ghost module to increase the diversity of features. The Ghost module generates more feature maps through a series of inexpensive operations. Specifically, the Ghost module first uses a 3×3 convolutional layer to generate a small number of "intrinsic" features $F_{\text{intrinsic}} \in \mathbb{R}^{H\times W\times k}$ from the reduced feature X', where $k << C/r$. Then, for each intrinsic feature $F_{\text{intrinsic},i}$, the Ghost module applies a series of simple linear transformations to generate more "Ghost" features $F_{\text{ghost},ij}$. These linear transformations are computationally inexpensive but can effectively increase the diversity of the features. The specific formula can be expressed as

$$F_{\text{ghost},ij} = \Phi_j\left(F_{\text{intrinsic},i}\right) \tag{16}$$

where $\Phi_j(\cdot)$ represents the jth linear transformation. Finally, the intrinsic features and Ghost features are combined to obtain the final feature map $F_{\text{ghost}} \in \mathbb{R}^{H\times W\times C/r}$. By combining the intrinsic features and Ghost features, we can obtain a feature map similar to that obtained using a standard convolutional layer, but at a much lower computational cost.

Furthermore, drop path is a regularization technique that helps prevent overfitting in neural networks. Applying drop path in the LPM module can further regularize the model and prevent overfitting.

To ensure that the dimension of the output is consistent with that of the input, we use another linear transformation layer implemented with a 1×1 convolution to restore the number of channels to the original dimension C. This operation can be expressed as

$$Y = \mathrm{Conv}_{1\times1}(F_{\text{se}}) \tag{17}$$

where F_{se} represents the feature tensor processed by the channel attention mechanism [30], and $Y \in \mathbb{R}^{H\times W\times C}$ is the final output feature tensor.

Through linear dimensionality reduction and expansion transformations, Ghost convolution, and channel attention mechanisms, the LPM module effectively reduces the number of parameters and computational complexity of the Transformer branch while ensuring feature representation ability.

E. Loss Function

To optimize the training of the DCTC-Net, we employ three loss functions: the overall loss of the DCTC-Net (L_{DCTC}), the loss of the CNN branch (L_{CNN}), and the loss of the Transformer branch (L_{Trans}). These three loss functions are defined as follows:

$$L_{\text{DCTC}} = L_{\text{mse}}\left(y_i^{\text{DCTC}}, y_i^{\text{label}}\right) + L_{\text{Dice}}\left(y_i^{\text{DCTC}}, y_i^{\text{label}}\right) \tag{18}$$

$$L_{\text{CNN}} = L_{\text{mse}}\left(y_i^{\text{CNN}}, y_i^{\text{label}}\right) + L_{\text{Dice}}\left(y_i^{\text{CNN}}, y_i^{\text{label}}\right) \tag{19}$$

$$L_{\text{Trans}} = L_{\text{mse}}\left(y_i^{\text{Trans}}, y_i^{\text{label}}\right) + L_{\text{Dice}}\left(y_i^{\text{Trans}}, y_i^{\text{label}}\right) \tag{20}$$

where $L_{\text{mse}}(\cdot)$ denotes the mean squared error loss, which measures the pixel-wise difference; and $L_{\text{Dice}}(\cdot)$ denotes the dice loss, which is used to better handle class imbalance. y_i^{DCTC}, y_i^{CNN}, y_i^{Trans}, and y_i^{label} represent the predicted output of the DCTC-Net, the predicted output of the CNN branch, the predicted output of the Transformer branch, and the ground truth label, respectively.

To balance the weights among these three loss functions, we adopt a Gaussian ramp-up curve to dynamically adjust them, which is defined as follows:

$$\lambda = e^{-5(1-k/K)^2} \tag{21}$$

where λ is a learnable weight coefficient, used to balance the fusion loss and the branch losses; k represents the current training epoch number; and K represents the total number of training epochs. This formula ensures that each branch independently learns effective features in the early stages of training. As training progresses, the fused L_{DCTC} loss guides the network to learn complementary information between branches, thereby improving the overall consistency and accuracy of segmentation.

The final overall loss function can be expressed as

$$L_{\text{Total}} = \lambda L_{\text{DCTC}} + ((1-\lambda)/2) L_{\text{CNN}} + ((1-\lambda)/2) L_{\text{Trans}}. \tag{22}$$

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To ensure that each branch can fully explore the feature space in the early stages of training, we initialize λ to 0.1, ensuring that the L_{CNN} and L_{Trans} losses dominate during the initial training phase, guiding each branch to learn features independently. As training progresses, the Gaussian ramp-up curve precisely controls the growth of the λ value, guiding the network to learn complementary information between branches, thereby improving the overall consistency and accuracy of segmentation.

F. Architecture Variants

We have built a DCTC-Net-T as a base network with a model size of 11.58 M and a computational cost of 4.53 GFLOPs. In addition, we built the DCTC-Net-B network to make a fair comparison with the latest networks, such as CvT [36] and PVT [37]. The window size is set to 7, and the input image size is 224×224. Other network parameters are set as follows:

1) *DCTC-Net-T:* Layer number = {2, 2, 6, 2, 6, 2, 2}, H = {3, 6, 12, 24, 12, 6, 3}, and D = 96, and
2) *DCTC-Net-B:* Layer number = {2, 2, 18, 2, 18, 2, 2}, H = {4, 8, 16, 32, 16, 8, 4}, and D = 96,

where D represents the number of image channels when entering the first layer of the dynamically adaptive CNNs branch and the cross-dimensional fusion Transformer branch, Layer number represents the number of Transformer blocks used in each stage, and H represents the number of multiple heads in SA. The specific details about different versions of DCTC-Net are shown in Table I.

IV. Experiment and Results

A. Datasets

We conducted research on the skin lesion segmentation dataset (ISIC2018) from the International Symposium on Biomedical Imaging (ISBI) [49], the Liver Tumor Segmentation (LiTS) Challenge dataset from the Medical Image Computing and Computer Assisted Intervention (MICCAI) Society [50], and the Automated Cardiac Diagnosis Challenge (ACDC) dataset from the University Hospital of Dijon (France) [51]. These three datasets have different data types and data distributions. The ISIC2018 dataset has electron microscope images, which contain 2594 dermoscopic images for training, but the ground truth images of the testing set have not been released, so we performed a fivefold cross-validation on the training set for a fair comparison. The LiTS dataset is a CT image of the human abdomen, containing 131 3-D CT liver scans, of which 100 scans are used for training, and the remaining 31 scans are used for testing. The ACDC dataset is an MRI image of the human heart, which contains cardiac short-axis Cine MRI data from 100 patients. The ACDC dataset includes healthy patients and patients with previous myocardial infarction, dilated cardiomyopathy, hypertrophic cardiomyopathy, and abnormal right ventricle (20 scans for each group). These data are obtained over a period of six years using two MRI scanners of two magnetic strengths (1.5 and 3.0 T). In addition, all images are empirically resized to 224×224 for efficiency.

B. Implementation Details and Evaluation Metrics

All networks are implemented using PyTorch 1.7 on one NVIDIA GeForce RTX 3090 GPU with 24 GB memory. For optimization, we used the Adam optimizer with an initial learning rate of 0.001. The learning rate is dynamically adjusted during training: it is halved if the validation loss does not decrease for ten consecutive epochs. To train the networks, we employed a combination of mean squared error (mse) loss and dice loss. As for other hyperparameter settings, we primarily followed the settings of mainstream methods in the field of medical image segmentation, while adapting them to our DCTC-Net and datasets. This is based on reports that CPFNet [33] and Swin-Unet [12] work well with such settings. Specifically, our total epoch number is set to 200.

In the experiment on the ISIC2018 dataset, we conducted an overall evaluation of mainstream medical image segmentation networks using five metrics: Dice (DI), Jaccard (JA), Sensitivity (SE), Accuracy (AC), and Specificity (SP). In the experiment on the LiTS-Liver dataset, we conducted an overall evaluation of mainstream medical image segmentation networks using five metrics: DI, VOE, RVD, ASD, and RMSD. In the experiment on the ACDC dataset, we conducted an overall evaluation of mainstream medical image segmentation networks using DI and 95HD [52].

The values of DI, JA, AC, SE, and SP range from 0 to 100, and better segmentation results mean that the values of evaluation metrics such as DI, JA, AC, SE, and SP will be higher, while the values of evaluation metrics such as VOE, RVD, ASD, RMSD, and 95HD will be lower [53]. These evaluation metrics are defined as

$$\text{DI} = \frac{2\text{TP}}{\text{FP} + 2\text{TP} + \text{FN}} \tag{23}$$

$$\text{JA} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{24}$$

$$\text{SE} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{25}$$

$$\text{AC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{26}$$

$$\text{SP} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{27}$$

where TP, TN, FP, and FN indicate the number of true positives, true negatives, false positives, and false negatives, respectively,

$$\text{VOE} = \left(1 - \left|\frac{A \cap B}{A \cup B}\right|\right) \times 100\% \tag{28}$$

$$\text{RVD} = \frac{|A| - |B|}{|B|} \times 100\% \tag{29}$$

$$\text{HD} = \max\left\{\sum_{a \in S(A)} \min_{b \in S(B)} \|a - b\|, \sum_{b \in S(B)} \min_{a \in S(A)} \|b - a\|\right\} \tag{30}$$

$$\text{ASD} = \frac{1}{|S(A)| + |S(B)|} \times \left(\sum_{a \in S(A)} \min_{b \in S(B)} \|a - b\| + \sum_{b \in S(B)} \min_{a \in S(A)} \|b - a\|\right) \tag{31}$$

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$$\text{RMSD} = \frac{1}{|S(A)| + |S(B)|} \times \sqrt{\sum_{a \in S(A)} \min_{b \in S(B)} \|a - b\|^2 + \sum_{b \in S(B)} \min_{a \in S(A)} \|b - a\|^2} \tag{32}$$

where A and B denote the ground truth and the segmentation result, respectively. $S(A)$ and $S(B)$ denote the set of surface voxels corresponding to A and B, respectively. In addition, the 95HD is defined as the 95th quantile of Hausdorff distances (HD) instead of the maximum [54].

C. Evaluation and Results

In this article, we conducted a comprehensive comparison across three multimodal datasets (ISIC2018, LiTS-Liver, and ACDC) between the proposed DCTC-Net and a range of mainstream medical image segmentation networks, including U-Net [1], Attention Unet [31], Swin-Unet [12], BRAU-Net++ [39], VM-Unet [45], and TMU [48], which possess inherent encoder–decoder structures. To ensure a fair comparison with CvT [36], PVT [37], and CrossForm [38], architectures lacking the decoder component, we augmented them with a decoder mirroring the structure of DCTC-Net's decoder to enable fair segmentation performance comparisons. Beyond presenting quantitative results, we provide a detailed analysis of the factors contributing to DCTC-Net's performance.

Table II shows the quantitative comparison between the proposed DCTC-Net and current mainstream CNNs, Transformer, and Mamba networks on the ISIC2018 dataset. From the experimental results, we conclude that our DCTC-Net achieves an optimal balance between model size and performance: it achieves the highest dice score of 91.23% without pretraining, while DCTC-Net-T delivers the lowest computational cost at just 4.53 GFLOPs and 11.58 M parameters, achieving a competitive dice score of 90.72%. While U-Mamba demonstrates some effectiveness, it achieves a lower dice score of 87.54% with a high computational cost of 453.89 GFLOPs. Our DCTC-Net-B network outperforms BAT, CvT, and CrossForm, with dice scores 1.02%, 3.00%, and 3.79% higher, respectively, while maintaining a lower parameter count. These superior results can be attributed to DCTC-Net's ability to address the specific challenges posed by dermoscopic images: blurred lesion boundaries, complex textures, and interfering factors such as hair and blood vessels. The DDConv module, with its ability to adaptively adjust the shape and weights of convolution kernels, is better equipped to capture the subtle structures and textures of lesion boundaries. Similarly, the (S)W-ACAM module effectively models the global contextual information in images, suppressing noise interference and helping the model to better distinguish between lesions and background.

Table III presents the quantitative comparison of DCTC-Net's performance on the LiTS-Liver dataset, alongside established mainstream segmentation networks. The results highlight DCTC-Net's superior performance in medical image segmentation, further confirming its ability to preserve both local and global features in medical images. Notably, DCTC-Net-B achieves SOTA performance with a dice score of 96.82 $\pm$ 1.22%, while DCTC-Net-T demonstrates a competitive second-best dice score of 96.48 $\pm$ 1.05%, both with considerably fewer parameters (21.24 M and 11.58 M, respectively) than several prominent methods. Furthermore, DCTC-Net-B demonstrates superior performance in VOE and all distance metrics compared to Mamba methods. These achievements can be attributed to the architecture's ability to address the unique challenges of liver segmentation: irregular liver and tumor shapes, low image contrast, and numerous small tumors. By adaptively adjusting the shape of its convolutional kernels, the DDConv module is particularly well-suited to segmenting the irregular contours of liver tumors. The dual-branch design, which fuses local and global information, provides greater robustness to noise and contrast variations, enabling the model to accurately infer tumor location and shape even in low-contrast regions.

Table IV shows the quantitative comparison of DCTC-Net's performance on the ACDC dataset alongside recent prominent medical image segmentation networks. The results demonstrate DCTC-Net's competitive performance in MRI-based multiorgan segmentation. DCTC-Net-B achieves an average DI score of 91.95 $\pm$ 0.66%, and the lowest average 95HD of 3.27 $\pm$ 0.51 mm. Although DCTC-Net-T and DCTC-Net-B do not achieve the best 95HD scores, they achieve superior average DI scores on this challenging dataset; significantly outperforming U-Mamba's dice score of 90.73 $\pm$ 0.76%. Notably, DCTC-Net's design specifically addresses the key challenges of cardiac MRI analysis, including dynamic motion, complex tissue structures, and interpatient variability. The dual-branch structure enables effective capture of both global cardiac anatomy and fine-grained tissue details, while the learned discriminative features better accommodate interpatient differences in heart morphology. The proposed method also outperforms other SOTA methods, such as the Transformer-based methods like MCAFT-Net (91.16 $\pm$ 0.79% dice score) and Mamba-based methods like DMA-Net (91.22 $\pm$ 0.71% dice score) used for comparison.

From the visualization in Fig. 6, it can be clearly observed that the DCTC-Net can preserve local detail features and global semantic features in medical images optimally. Therefore, it achieves accurate segmentation results for segmentation targets with irregular edges and significant scale variations. Fig. 6(a) shows the final output result of the CNN branch, and Fig. 6(b) shows the final output result of the Transformer branch. We can observe that the CNN branch has more accurate control over the local detail information of segmented targets. However, since CNN branches mainly capture local features of images, they are more susceptible to noise interference. In contrast, the Transformer branch has more precise control over target localization. Since the Transformer branch mainly captures the global features of the image, it easily overlooks the detailed information of the segmented target. Therefore, after integrating the CNN branch with the Transformer branch, the DCTC-Net fully leverages the structural and generalization advantages of CNNs and Transformers, providing better local and global feature representations in medical image data, demonstrating great potential in the field of medical image segmentation.

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TABLE IV

PERFORMANCE COMPARISON OF THE PROPOSED METHOD AGAINST THE SOTA APPROACHES ON THE ACDC BENCHMARKS. RED INDICATES THE BEST RESULT, AND BLUE DISPLAYS THE SECOND BEST

Model		RV		MYO		LV		AVG	
		DI	95HD	DI	95HD	DI	95HD	DI	95HD
CNNs	U-Net [1]	90.11±1.20	5.90±1.36	88.87±0.33	2.49±0.51	94.16±1.06	2.95±0.77	91.05±0.68	3.78±0.75
	R2UNet [17]	90.03±1.19	6.92±2.30	89.15±0.42	2.46±0.32	94.33±1.04	2.85±0.46	91.17±0.72	4.08±0.77
	Attention Unet [31]	90.41±1.06	5.48±0.93	88.94±0.36	2.81±0.78	94.14±1.37	3.26±1.32	91.16±0.82	3.85±0.93
	CENet [32]	89.77±0.28	5.32±0.29	88.95±0.23	2.53±0.28	94.16±1.33	2.93±1.12	90.96±0.22	3.59±0.81
	3D Unet [9]	90.25±0.82	4.95±1.02	89.07±0.41	2.55±0.33	94.23±1.40	3.17±1.09	91.18±0.47	3.56±0.69
	V-Net [10]	90.01±0.76	5.05±0.88	89.00±0.29	2.47±0.46	94.13±1.31	3.24±1.43	91.05±0.72	3.59±0.83
	SSNet [34]	90.13±1.02	5.34±0.95	89.06±0.54	2.46±0.57	94.27±1.24	2.89±1.12	91.15±0.81	3.57±0.62
Transformer	Swin-Unet † [12]	90.35±1.13	5.45±1.08	88.73±0.49	3.16±1.32	94.03±1.43	3.77±2.23	91.04±0.93	4.13±1.51
	TransUNet † [35]	89.79±1.28	6.35±1.31	88.77±0.56	3.09±1.30	94.05±1.27	3.38±1.55	90.90±0.98	4.27±1.26
	CvT † [36]	90.05±1.36	5.82±1.43	88.74±0.39	3.11±0.98	94.06±1.36	3.45±1.65	90.95±0.86	4.13±1.34
	PVT [37]	89.52±1.31	6.51±1.35	88.08±0.52	3.12±1.35	93.75±1.25	3.34±1.53	90.45±0.78	4.32±1.14
	CrossForm [38]	89.66±1.48	6.23±1.51	88.17±0.49	3.08±1.27	93.97±1.06	3.28±1.42	90.60±0.74	4.19±1.08
	BRAU-Net++ [39]	89.87±1.32	5.87±1.21	88.62±0.51	2.84±1.12	94.16±1.35	2.98±1.18	90.88±0.56	3.90±0.94
	HResFormer [40]	89.94±1.23	5.23±0.81	89.03±0.46	2.46±0.70	94.29±1.29	2.73±0.97	91.09±0.83	3.47±0.71
	MCAFT-Net [41]	89.57±0.95	5.28±0.68	89.37±0.49	2.49±0.62	**94.55±1.31**	**2.61±0.79**	91.16±0.79	3.46±0.69
	CSWin-UNet [42]	89.19±0.88	**4.83±0.45**	**89.63±0.43**	2.43±0.48	94.26±1.18	2.65±0.81	91.02±0.63	**3.31±0.49**
Mamba	U-Mamba [43]	89.63±1.53	5.93±1.32	88.48±0.43	2.86±0.97	94.08±1.27	3.05±1.24	90.73±0.76	3.94±0.78
	Mamba-UNet [44]	89.75±1.56	5.76±1.14	88.53±0.59	2.77±1.14	93.85±1.24	2.96±1.16	90.71±0.81	3.83±0.72
	VM-Unet [45]	89.78±1.18	5.64±1.10	88.76±0.41	2.65±1.05	94.28±1.31	2.93±1.02	90.94±0.68	3.74±0.64
	UD-Mamba [46]	89.89±1.35	5.69±1.16	88.93±0.44	2.68±0.76	94.26±1.28	2.95±1.09	91.03±0.74	3.77±0.75
	DMA-Net [47]	90.02±1.26	5.48±0.98	89.31±0.38	**2.41±0.87**	94.32±1.25	2.91±0.85	91.22±0.71	3.60±0.61
	TMU † [48]	89.97±1.16	5.34±0.67	88.74±0.47	2.45±0.92	94.22±1.17	2.89±0.77	90.97±0.69	3.56±0.68
DCTC-Net-T (our)		**90.46±1.15**	5.27±0.55	89.33±0.48	2.43±0.46	94.38±1.23	2.87±0.81	**91.39±0.82**	3.53±0.54
DCTC-Net-B (our)		**90.95±0.65**	**4.86±0.37**	**89.62±0.36**	**2.32±0.82**	**95.30±1.08**	**2.63±0.83**	**91.95±0.66**	**3.27±0.51**

† indicates that the model is initialized with pre-trained weights on ImageNet21K.

D. Failure Case Analysis

While DCTC-Net demonstrates robust performance across various datasets, analyzing its limitations is essential for guiding future research. We present typical failure cases in Fig. 7 to provide a comprehensive evaluation.

Fig. 7 illustrates two primary failure modes. The first two rows show instances of inadequate handling of blurred boundaries. In these examples, the target objects exhibit extremely low contrast with surrounding tissue, and boundaries are inherently ambiguous. Our DCTC-Net, while attempting to delineate regions, produces contours that are either discontinuous or slightly inaccurate. This highlights a challenge in scenarios with very low signal-to-noise ratios, where precise edge determination is difficult even for human experts. Future work could address this by incorporating advanced boundary-aware mechanisms, such as explicit boundary prediction branches during training or loss functions that heavily penalize boundary errors.

The bottom two rows of Fig. 7 display cases of missed small target detection. Here, DCTC-Net fails to identify very small-scale lesions. A likely cause is the hierarchical downsampling process in the encoder. During successive pooling or strided convolution operations, feature information corresponding to tiny targets can be significantly diluted or lost entirely, making accurate decoder reconstruction challenging. To mitigate this issue, future improvements could focus on optimizing network architecture to better preserve small-scale features, for example, by modifying downsampling strategies, enhancing cross-scale feature fusion using techniques like feature pyramid networks (FPNs), or designing loss functions that dynamically increase weights for small, hard-to-detect objects.

By acknowledging and analyzing these failure cases, we identify clear pathways for enhancing DCTC-Net's robustness and accuracy in future work.

TABLE V

ABLATION EXPERIMENTS OF DDCONV, (S)W-ACAM, AND LPM IN DCTC-NET IN THE ISIC2018 DATASET

Backbone	DDConv	(S)W-ACAM	LPM	Para. (M)	DI (%) ↑
U-Net+Swin-Unet				46.92	87.45
U-Net+Swin-Unet	√			48.25	89.15
U-Net+Swin-Unet		√		30.26	89.62
U-Net+Swin-Unet			√	15.45	88.43
U-Net+Swin-Unet	√	√		32.16	90.88
U-Net+Swin-Unet	√		√	16.93	89.12
U-Net+Swin-Unet		√	√	9.67	89.46
U-Net+Swin-Unet	√	C-C	√	10.69	88.82
U-Net+Swin-Unet	√	C-H	√	10.24	89.05
U-Net+Swin-Unet	√	C-W	√	10.24	89.06
U-Net+Swin-Unet	√	H-W	√	10.05	88.95
DCTC-Net-T (our)	√	√	√	11.58	90.72

E. Ablation Study

To fully demonstrate the effectiveness of the different modules within DCTC-Net, we conducted a series of ablation experiments on the ISIC2018 dataset. Table V summarizes the results and demonstrates that the DDConv, (S)W-ACAM, and LPM modules each contribute significantly to the model's performance.

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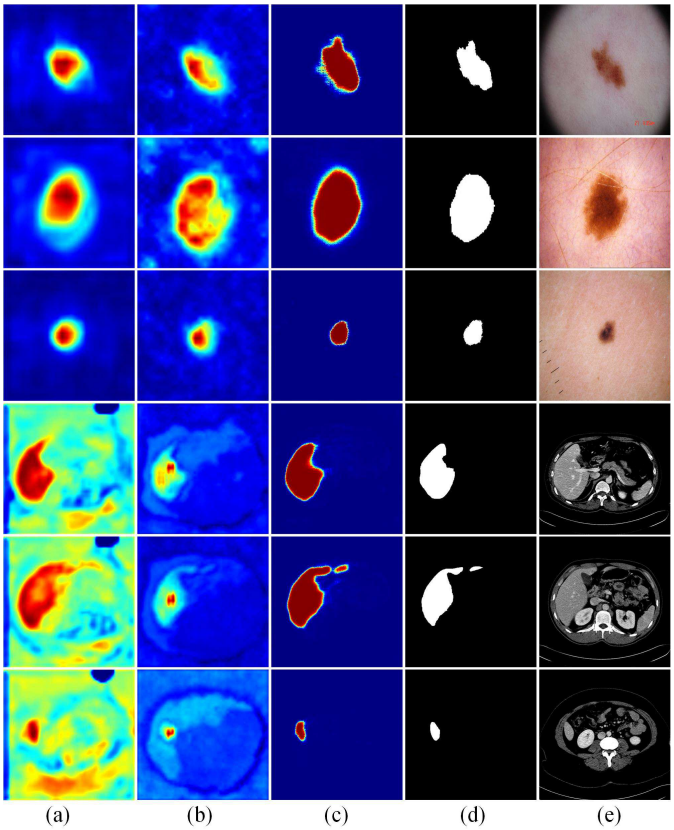


Fig. 6. Visualization results of DCTC-Net on ISIC2018 dataset and LiTS-Liver dataset. Predicted image output by (a) CNNs branch, (b) Transformer branch, and (c) DCTC-Net. (d) Corresponding label. (e) Corresponding original image.

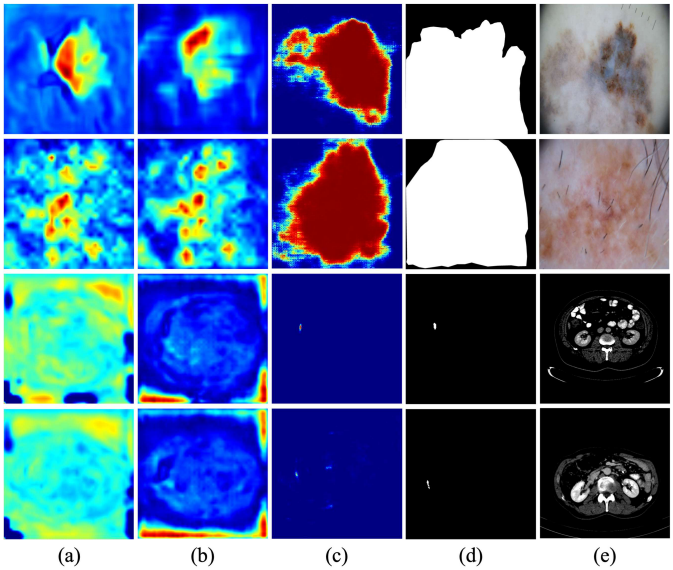


Fig. 7. Visualization of typical failure cases. The first two rows show poor handling of blurred boundaries. The last two rows show missed detection of small targets. Predicted image output by (a) CNNs branch, (b) Transformer branch, and (c) DCTC-Net. (d) Corresponding label. (e) Corresponding original image.

The addition of the DDConv module to the U-Net+Swin-Unet baseline results in an increase in the dice coefficient from 87.45% to 89.15%. This indicates that the DDConv module is effective in capturing the fine-grained structures and texture information of lesion boundaries in dermoscopic images, thereby improving segmentation accuracy. The adaptive nature of DDConv enables it to better handle cases where lesion edges are blurred, contributing to the improvement of the model's accuracy. Table V shows that the addition of the DDConv module improves the overall dice score by nearly 2%.

Adding the (S)W-ACAM module to the U-Net+Swin-Unet baseline leads to an increase in the dice coefficient from 87.45% to 89.62%, while reducing the number of parameters by 16.66 M. This suggests that the (S)W-ACAM module is effective in modeling the global contextual information of the image, suppressing noise interference, and helping the model better distinguish between lesions and the background. This global context modeling ability is crucial for reducing the impact of interfering factors such as hair and blood vessels. Table V also demonstrates the importance of cross-dimensional feature extraction. When (S)W-ACAM is replaced with cross-channel (C-C) attention mechanisms, cross-height (C-H) attention mechanisms, cross-width (C-W) attention mechanisms, and height–width (H-W) attention mechanisms, the results show lower dice scores. This demonstrates that the cross-dimensional feature extraction capability of (S)W-ACAM is key for performance. From the ablation results, it can be seen that (S)W-ACAM improves the final model accuracy, while significantly reducing the number of model parameters, contributing to model efficiency.

Table V shows that the number of parameters is greatly reduced after using the LPM module. This shows that the LPM module can significantly reduce the number of model parameters while maintaining model performance. However, the model performs modestly when using LPM alone, so it needs to be combined with the (S)W-ACAM module to achieve better performance.

In summary, the ablation experiments clearly demonstrate that the DDConv, (S)W-ACAM, and LPM modules each play crucial roles in DCTC-Net, and that a synergistic effect exists between them. DDConv effectively captures local detailed features, (S)W-ACAM effectively models global contextual information, and LPM significantly reduces the number of model parameters while maintaining model performance. The effective combination of these modules enables DCTC-Net to achieve excellent results in medical image segmentation tasks.

F. Network Visualization

To facilitate understanding of the DCTC-Net, we visualized the feature maps at each stage of the network, as shown in Fig. 8. We use linear interpolation to upsample the deep feature maps, which initially have low spatial resolution, to the same size as the input and output images. In the visualization, red represents areas that the network focuses on more, while blue represents areas that the network focuses on less.

The visualization results demonstrate the entire process of feature extraction and fusion, as well as other transformations that the image undergoes in the network. In the two branches of the encoder, the model analyzes and identifies the semantic information in the image. In the two branches of the decoder,

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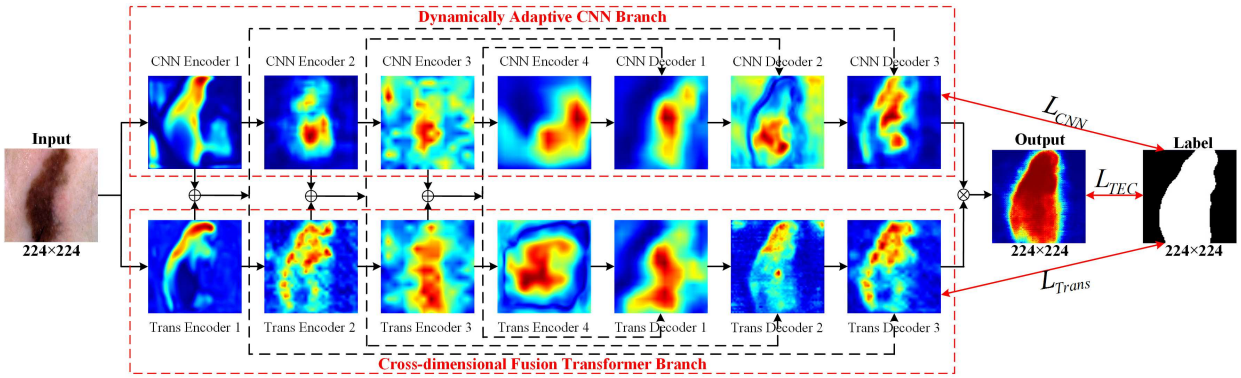


Fig. 8. Visualization of results at each layer of DCTC-Net—taking ISIC2018 dataset image segmentation as an example.

after skip connections and feature fusion operations, the model begins to focus more on the semantic features of the target region. The information exchange and fusion between the CNN and Transformer branches play an important role in the accurate segmentation of the target region.

Overall, as the network depth increases, the DCTC-Net gradually refines the localization and contour segmentation of semantic objects in medical images, demonstrating the effectiveness of our proposed DCTC-Net for global information modeling and accurate segmentation of medical images.

V. Discussion

Despite exhibiting promising performance in medical image segmentation tasks, DCTC-Net, like all deep learning models, faces certain limitations in specific application scenarios. To comprehensively evaluate DCTC-Net's applicability and guide future research, we examine its limitations concerning data dependence, computational resource requirements, and generalization ability.

First, our DCTC-Net relies on a substantial amount of labeled data for training. While techniques such as the Gaussian ramp-up curve are employed to mitigate the small sample problem, DCTC-Net may still face the risk of overfitting when available labeled data is severely limited, leading to degraded generalization on unseen data. This limitation is particularly prominent in scenarios involving rare diseases or novel medical imaging modalities, where obtaining sufficient labeled data is often challenging.

Second, the dual-branch structure and use of Transformer modules in DCTC-Net contribute to a relatively high parameter count and computational complexity. Although we have optimized the model's efficiency, deploying DCTC-Net and performing real-time inference may still present challenges to devices with limited computational resources, such as embedded medical devices or mobile diagnostic platforms. Therefore, further reducing DCTC-Net's computational complexity in resource-constrained scenarios is a noteworthy research direction.

Third, DCTC-Net's performance is highly dependent on the distribution of the training data. We have primarily validated the effectiveness of DCTC-Net on datasets such as ISIC2018, LiTS-Liver, and ACDC, which encompass various medical imaging modalities such as dermoscopic images, CT images, and MRI images. However, DCTC-Net's performance requires further validation for other types of medical images. Furthermore, domain shift, arising from differences in imaging parameters, image contrast, and noise levels, may exist between different datasets. Enhancing DCTC-Net's generalization ability across various data types and domains for future research.

Future research can focus on several aspects to enhance DCTC-Net's applicability and robustness. First, weakly supervised learning methods, such as semi-supervised learning [55], [56] and self-supervised learning [57], could be explored to reduce the reliance on labeled data. Second, techniques like model compression [58] and knowledge distillation [59] can be employed to further reduce computational complexity. Third, domain adaptation learning methods [60] should be investigated to improve generalization across different medical image types and domains.

We believe that our DCTC-Net will play a crucial role in the field of medical image segmentation. Through continuous exploration and improvement, it will provide enhanced support for clinical diagnosis and treatment.

VI. Conclusion

In this study, we proposed a novel architecture, DCTC-Net, which combines dynamically adaptive CNNs and a cross-dimensional fusion Transformer in a parallel configuration for medical image segmentation. The proposed DCTC-Net integrates the strengths of both CNNs and Transformers, preserving both local details and global semantic features of medical images by leveraging local relationship modeling and long-range dependence modeling. The proposed DDConv overcomes the limitations of fixed receptive fields and parameter sharing inherent in vanilla convolution, improving local feature representation and enabling adaptive extraction of spatial features. The proposed (S)W-ACAM SA mechanism comprehensively captures the cross-dimensional correlation between spatial features and channels, and adaptively learns salient information between spatial and channel dimensions through network training. Furthermore, by replacing the MLP in traditional Transformers with LPM, DCTC-Net significantly reduces the number of parameters, reduces dependence on

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pretraining, addresses data scarcity challenges in labeled medical images, and reduces over-fitting. Compared to existing CNN and Transformer-based medical image segmentation networks, our DCTC-Net exhibits significant advantages in both operational efficiency and segmentation effectiveness.

While DCTC-Net has demonstrated encouraging results, several directions warrant further exploration. First, although the model performs well across multiple modalities, its generalization on broader medical image types, such as ultrasound and digital pathology, remains to be validated; incorporating domain adaptation techniques will be crucial. Second, to meet deployment demands on extremely resource-constrained devices, advanced model compression techniques like network pruning and quantization could be explored to optimize efficiency further. Finally, integrating task-specific prior knowledge, such as anatomical shape constraints, could enhance segmentation robustness in complex cases and improve model interpretability, making it more valuable for clinical applications.

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